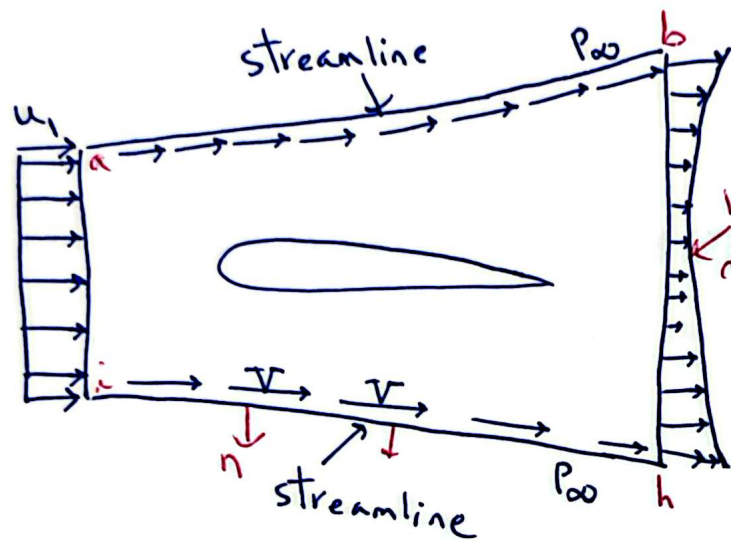
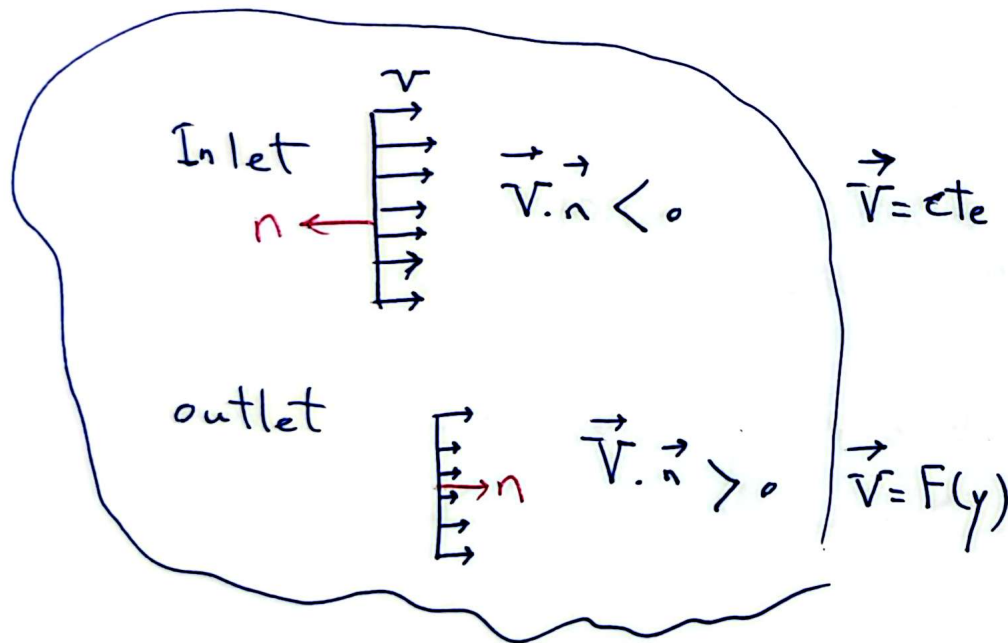




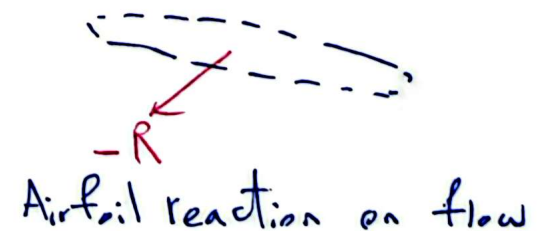
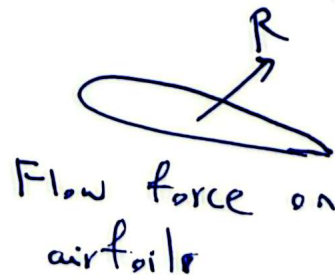
streamline is a line tangent to velocity field!

velocity deficit $\vec{V} \cdot \vec{n} = 0 \rightarrow$ no mass transfer across the streamline

over the airfoil, boundary layer forms and we have velocity deficit in some regions, so to keep mass conserved, area increases (streamlines diverge)



We can remove airfoil and put its resultant force instead



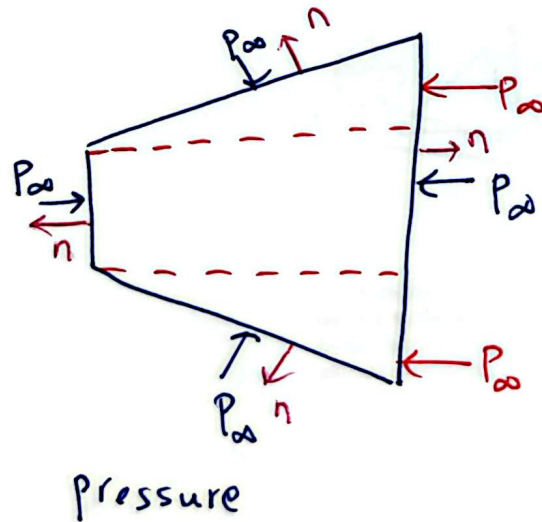
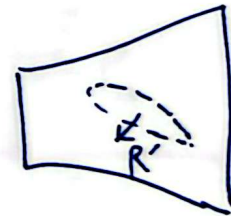
Forces are pressure force and $-R'$

$$\Rightarrow F = - \int_{abhi} p ds - R' \quad \text{ignore body force}$$

write Mom. Eqⁿ in integral format

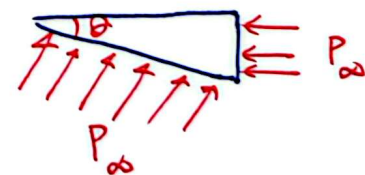
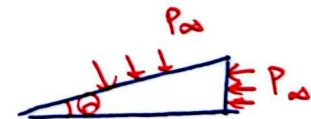
$$\frac{\partial}{\partial t} \iiint \rho V dV + \iint_S (\rho V \cdot ds) V = - \underbrace{\iint p ds}_{\text{force}} - R'$$

Steady flow



$$\iint_S P_{\infty} ds = 0$$

$$\iint P_{\infty} dA = .$$





$$\vec{F}_x = P_\infty \sin \theta (L) = P_\infty (\underbrace{L \sin \theta}_A) = \vec{P}_\infty A \quad \text{to the right}$$

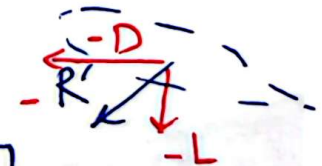


$$P_\infty L \sin \theta = P_\infty A \longrightarrow \text{Pressure force cancels!}$$

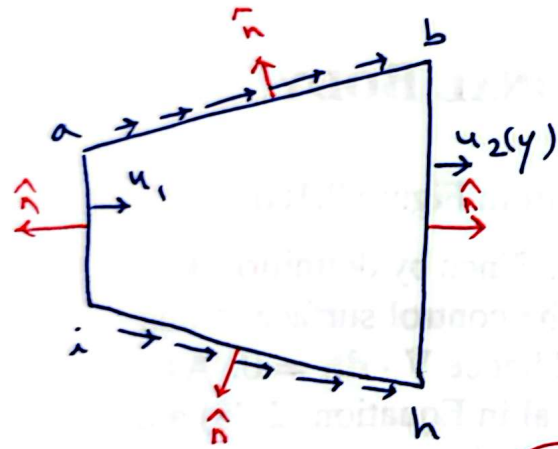
$$\implies \iint_S P_\infty dA = 0 \quad \text{if } P = P_\infty \text{ on all surfaces!}$$

$$\text{So Mom: } \iint (\rho \vec{V} \cdot d\vec{s}) \vec{V} = -R'$$

$$\boxed{x\text{-Mom: } \iint (\rho \vec{V} \cdot d\vec{s}) u = -R'_x = -D}$$



so $D = - \iint (\rho \underbrace{\vec{V} \cdot d\vec{s}}_{|\vec{s}| \hat{n}}) u$, to compute drag, compute the integral over four surfaces!



$\vec{V} \cdot d\vec{s} =$ over streamlines $\begin{cases} a-b \\ i-h \end{cases}$

$$\vec{V} \cdot d\vec{s} = \vec{V} \cdot \hat{n} ds \quad \text{as } \vec{V} \cdot \hat{n} = 0$$

on $a-i$ $\vec{V} \cdot d\vec{s} < 0$

on $b-h$ $\vec{V} \cdot d\vec{s} > 0$

$$-D = - \int_i^a \rho u_1 u_1 dy + \int_h^b \rho u_2(y) u_2(y) dy + 0 + 0$$

$ds = dy \times 1$ (depth is unity)

If $u_2(y)$ profile is given, compute the above integral.

let's write continuity Eq.:

$$\frac{\partial}{\partial t} \iiint \rho d(\text{Vol}) + \iint \rho \vec{V} \cdot d\vec{s} = 0 \Rightarrow \text{continuity: } \iint \rho \vec{V} \cdot d\vec{s} = 0$$

$\overset{\text{steady}}{\rho}$

$$-\int_i^a \rho u_1 dy + \int_h^b \rho u_2(y) dy + 0 + 0 = 0$$

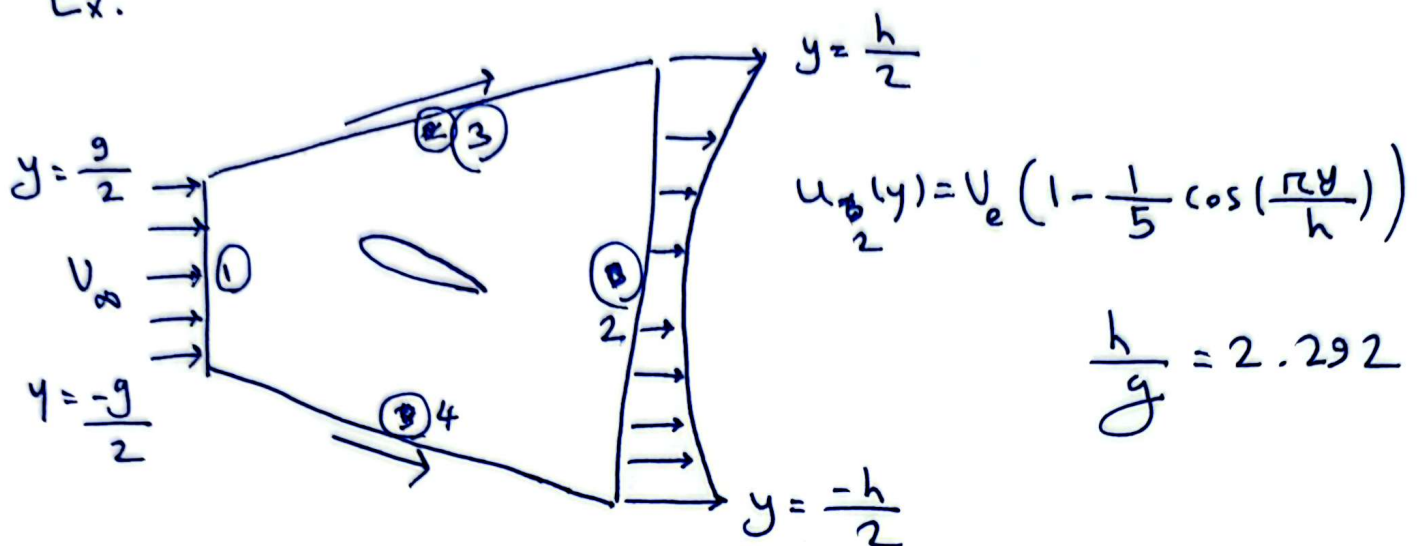
$$\Rightarrow \int_a^b g u_1 dy = \int_h^b g u_2(y) dy \quad \times u_1$$

$$\Rightarrow \boxed{\int_a^b g u_1^2 dy = \int_h^b g u_2(y) u_1 dy} \rightarrow \text{put in Mom. (D-Eq)}^{\text{a}}$$

$$\Rightarrow -D = -\int_h^b g u_2(y) u_1 dy + \int_h^b g u_2(y) u_2(y) dy = \int_h^b g u_2(y) (u_2(y) - u_1) dy$$

$$\Rightarrow \boxed{D = \int_h^b g u_2(y) (u_1 - u_2(y)) dy}$$

Ex.



compute D as a function of ρ_∞ , U_∞ and g

Solution:

$$D = - \iint (\rho \mathbf{V} \cdot d\mathbf{s}) u$$

$$D = - \left[\int_{-\frac{h}{2}}^{\frac{h}{2}} \rho u_2^2(y) dy - \int_{-\frac{g}{2}}^{\frac{g}{2}} \rho u_1 u_1 dy \right] \quad (1)$$

But note that integration of $u_2^2(y)$ is a bit difficult, so, use the other equation for drag!

$$D = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho u_2(y) (u_1 - u_2(y)) dy = - \int_{-\frac{g}{2}}^{\frac{g}{2}} \rho u_1 (u_2 - u_1) dy \quad (2) \quad (3)$$

(1) = (2) = (3) All are correct!

(1)

part ① $D_1 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho u_2(y) u_1 = \rho u_e \left(1 - \frac{1}{5} \cos\left(\frac{\pi y}{h}\right)\right) u_1$

part ② $D_2 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho u_2^2(y) = -\rho U_e^2 \left[1 - \frac{1}{5} \cos\left(\frac{\pi y}{h}\right)\right]^2$

$D = 0.245 \rho U_\infty^2 h \rightarrow$ This solution used $U_e = \frac{U_\infty}{2}$

let's prove it:

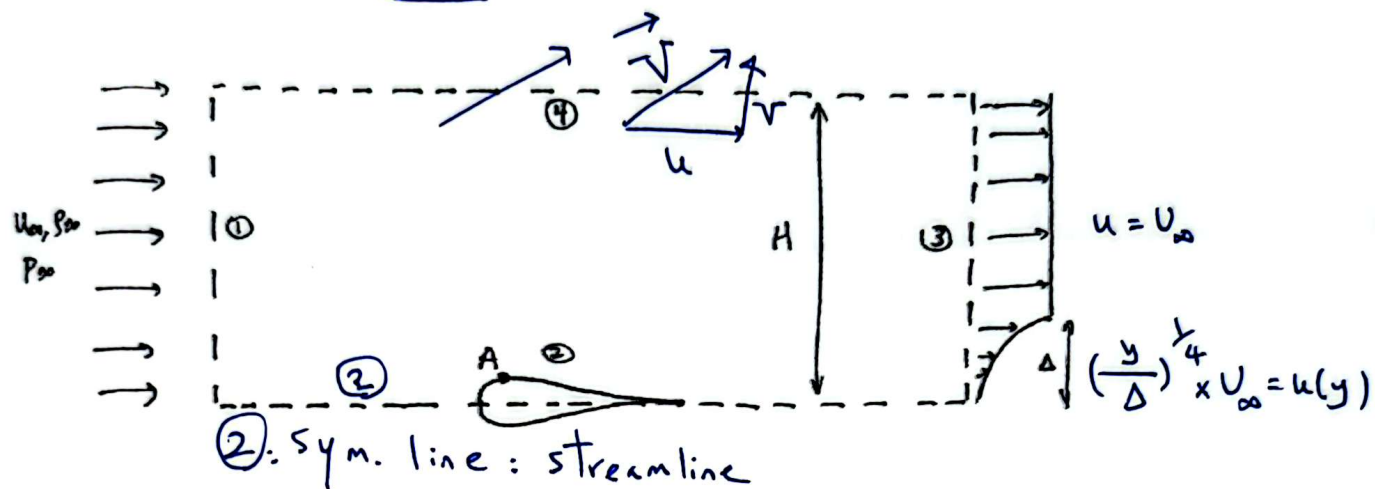
Continuity : $\int_{-\frac{g}{2}}^{\frac{g}{2}} \rho V \cdot ds + \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho V \cdot ds = 0$

$\int_{-\frac{g}{2}}^{\frac{g}{2}} -\rho U_\infty dy + \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho U_e \left(1 - \frac{1}{5} \cos\left(\frac{\pi y}{h}\right)\right) dy = 0, \frac{h}{g} = 2.292$

$-\cancel{\rho} U_\infty g + \cancel{\rho} \left[U_e h - \frac{2h}{\pi} \frac{U_e}{5} \right] = 0$

$U_\infty g = U_e h \left[1 - \frac{2}{\pi 5}\right] \Rightarrow U_e = \frac{U_\infty \left(\frac{g}{h}\right)}{0.873} = \frac{U_\infty}{2}$

2- The velocity profile of the flow going over the surface of an airfoil in region 3 is measured as follows: ($\Delta = 0.04m$).



- Calculate the mass flow rate passing through surface 4 using the continuity equation.
- Calculate the drag force on the airfoil using the momentum and continuity equations. (Note that surface 2 (the lower boundary of the control volume) is a line of symmetry., and the wake shape is a ~~quarter of a circle~~)

$$u_1 = U_\infty$$

$$u_3 = \begin{cases} U_\infty \left(\frac{y}{\Delta}\right)^{\frac{1}{4}} & 0 < y < \Delta \\ U_\infty & \Delta < y < H \end{cases}$$

$$\dot{m}_{(4)} = \dot{m}_{(1)} - \dot{m}_{(3)} = \int_{-\infty}^{\infty} U_\infty H - \left[\int_0^{\Delta=0.04} \int_{-\infty}^{\infty} U_\infty \left(\frac{y}{\Delta}\right)^{\frac{1}{4}} dy + \int_{\Delta}^H \int_{-\infty}^{\infty} U_\infty dy \right]$$

$$\dot{m}_{(4)} = \cancel{\int_{-\infty}^{\infty} U_\infty H} - 0.035 \int_{-\infty}^{\infty} U_\infty + \cancel{\int_{-\infty}^{\infty} U_\infty (H - 0.04)}$$

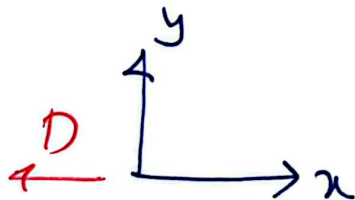
$$\boxed{\dot{m}_4 = 0.005 \int_{-\infty}^{\infty} U_\infty} \text{ going out of } (4)$$

$$D = - \underbrace{\int_0^H \rho_{\infty} u_1^2 dy} + \int_0^H \rho_{\infty} u_2^2 dy$$

$$D = - \rho_{\infty} U_{\infty}^2 H + \underbrace{\int_0^{\Delta} \rho_{\infty} U_{\infty}^2 \left(\frac{y}{\Delta}\right)^{\frac{2}{4}} dy}_{\Delta} + \underbrace{\int_{\Delta}^H \rho_{\infty} U_{\infty}^2 dy}_{\Delta} =$$

$$D = - \cancel{\rho_{\infty} U_{\infty}^2 H} + \rho_{\infty} U_{\infty}^2 (0.031) + \cancel{\rho_{\infty} U_{\infty}^2 (H - \Delta)}$$

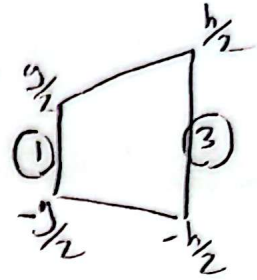
$$D = \rho_{\infty} U_{\infty}^2 (-\Delta + 0.031) = \boxed{-0.009 \rho_{\infty} U_{\infty}^2}$$



$$D = - \left[\iint_S \mathbf{V} \cdot d\mathbf{s} u \right]$$

$$D = - \left[\int_{-h/2}^{h/2} \int_{-g/2}^{g/2} \mathbf{V} \cdot d\mathbf{s} u - \int_{-g/2}^{g/2} \int_{-h/2}^{h/2} \mathbf{V} \cdot d\mathbf{s} u \right]$$

$$D = - \left[\int_{-h/2}^{h/2} \int_{-g/2}^{g/2} \rho u_3^2 dy - \int_{-g/2}^{g/2} \int_{-h/2}^{h/2} \rho u_1^2 dy \right] \quad \text{Eq. (1) for the drag}$$



Continuity $\left[\int_{-g/2}^{g/2} \rho u_1 dy = \int_{-h/2}^{h/2} \rho u_3 dy \right] \times u_1$

$$\left[\int_{-g/2}^{g/2} \rho u_1^2 dy = \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] \quad (2)$$

replace $\int_{-g/2}^{g/2} \rho u_1^2 dy$ from (2) in (1), we get:

$$D = - \left[\int_{-h/2}^{h/2} \rho u_3^2 dy - \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] = - \left[\int_{-h/2}^{h/2} \rho u_3 (u_3 - u_1) dy \right]$$

Eq. 3 for drag

multiply continuity by u_3 , you get:

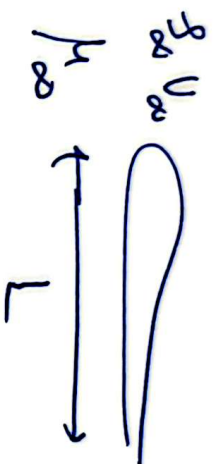
$$\int_{-g/2}^{g/2} \rho u_1 u_3 dy = \int_{-h/2}^{h/2} \rho u_3^2 dy \quad (4)$$

replace $\int_{-h/2}^{h/2} \rho u_3^2 dy$ from (4) in (1) we get:

$$D = - \left[\int_{-g/2}^{g/2} (\rho u_1 u_3 - \rho u_1^2) dy \right] = - \left[\int_{-g/2}^{g/2} \rho u_1 (u_3 - u_1) dy \right] \quad (5)$$

Eq. (1), (3) & (5) all give the same value for drag

$$Re = \frac{\rho_a V_\infty L}{\mu}$$



Reynolds number regimes

TABLE 2.2 Reynolds Number Regimes and Characteristics for an Airfoil

Reynolds number regime	Reynolds number range	Characteristics
Creep	$0 < Re_L < \approx 10^2$	Viscous effects dominate, drag inversely proportional to Re_L
Low Re_L laminar	$\approx 10^2 < Re_L < \approx 10^4$	Vortex shedding, separation bubbles, reduced lift, drag proportional to Re_L
Laminar	$\approx 10^4 < Re_L < \approx 10^5$	Separation bubbles, decreased skin friction, earlier separation, earlier stall
Transitional	$\approx 10^5 < Re_L < \approx 10^6$	Both laminar and turbulent boundary layers with characteristics of both
Turbulent	$Re_L > \approx 10^6$	Inertial effects dominate, increased skin friction, delayed separation, delayed stall

Figure 1.8: Table. 2.2 from AE-BC